BLW Surfaces and the Björling Problem

English summary of the Master's thesis
Mặt BLW và Bài toán Björling

Huỳnh Kỳ Anh

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Advisor: Assoc. Prof. Đoàn Thế Hiếu
with the assistance of Prof. José A. Gálvez

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Abstract

The classical *Björling problem*, posed by E. G. Björling in 1844 and solved by H. A. Schwarz in 1890, asks for the minimal surface in $\mathbb{R}^3$ containing a prescribed analytic curve with a prescribed unit normal field along it. This thesis poses and solves the analogous problem for the class of *BLW surfaces* ("Linear Weingarten surfaces of Bryant type") in hyperbolic 3-space $\mathbb{H}^3$: surfaces whose mean curvature H and (extrinsic) Gauss curvature K_I satisfy the linear relation $2a(H-1)+bK_I = 0$. This class contains the `CMC1` surfaces (constant mean curvature one) of Gálvez–Mira [1] ($b = 0$) and the flat surfaces of Gálvez–Mira [3] ($a = 0$) as special cases. The main results are: (1) a full existence-and-uniqueness theorem for the Björling problem for BLW surfaces, together with an explicit closed-form representation of the solution (a "Björling representation" that does not go through the conformal/harmonic representation of [2]); and (2) a theorem determining exactly when the hyperbolic Gauss map of a BLW surface is meromorphic versus anti-meromorphic, generalizing R. Bryant's classical fact that the hyperbolic Gauss map of a regular surface in $\mathbb{H}^3$ is meromorphic iff the surface has $H \equiv 1$ (i.e., is `CMC1`).

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1 Introduction

The *classical Björling problem* was introduced by E. G. Björling in 1844 and solved by H. A. Schwarz in 1890: given an analytic curve $\beta \subset \mathbb{R}^3$ with an analytic unit tangent, find the minimal surface containing β as a geodesic. A key structural consequence of the existence-and-uniqueness of the solution is that a minimal surface Σ is symmetric with respect to any plane orthogonal to Σ.

A natural generalization is: given a class $\mathfrak{B}$ of surfaces (e.g. `CMC1`, `ELW`, `flat`, ...), a curve β, and a unit normal field V along β, find a surface $\Sigma \in \mathfrak{B}$ containing β with prescribed unit normal V along it. Appendix A of the thesis lists the Björling-type problems that had been solved before this work; in every known case the solution is unique and yields a symmetry principle for the surface class in question. Solving the Björling problem for a class also produces many explicit examples and has led to classification results — e.g. the classification of `CMC1` surfaces of cylindrical topology [1], and of flat fronts with embedded isolated singularities [3].

Building on the solutions for `CMC1` surfaces [1] and regular flat surfaces [3], this thesis poses and solves the Björling problem for **BLW surfaces**: surfaces $\psi : S \to \mathbb{H}^3$ whose mean curvature H and Gauss curvature $K_I = K - 1$ (here K is the Gauss–Kronecker/extrinsic curvature) satisfy

$$2a(H-1) + bK_I = 0, \tag{1}$$

for real constants a, b with $a + b \neq 0$. The technique follows [1]: to build the solution one needs a fact about the meromorphicity of the hyperbolic Gauss map, originally due to R. Bryant [5]. This fact is partially generalized in Theorem 11 below, which is the key ingredient for solving the Björling problem for BLW surfaces. (Open questions about further generalizing Bryant's theorem are discussed in Section 4, Problem 9..)

Outline

Section 2 collects the background material: the model of hyperbolic space, the local theory of surfaces in $\mathbb{H}^3$, Riemann surfaces and (anti-)meromorphic maps, conformality, basic lemmas on BLW surfaces, and the Cauchy problem for the Liouville equation. Section 3 states the Björling problem for BLW surfaces and the two main theorems (Theorem 10, the existence-and-uniqueness theorem, and Theorem 11, on meromorphicity of the hyperbolic Gauss map), and sketches their proofs. Section 4 summarizes the contributions and lists several open problems suggested by the work. Appendix A reproduces the thesis's table of previously solved Björling-type problems.

2 Preliminaries

2.1 The hyperbolic space $\mathbb{H}^3$

Lorentz 4-space $\mathbb{L}^4$ is $\mathbb{R}^4$ equipped with the Lorentzian pseudo-metric

$$\langle \cdot, \cdot \rangle = -dx_0^2 + dx_1^2 + dx_2^2 + dx_3^2. \tag{2}$$

Two distinguished subsets are the *de Sitter* 3*-space* $\mathbb{S}^3_1 = \{x \in \mathbb{L}^4 : \langle x, x\rangle = 1\}$ and the *future light cone* $\mathbb{N}^3_+ = \{x \in \mathbb{L}^4 : \langle x, x\rangle = 0,\ x_0 > 0\}$. The set

$$\mathbb{H}^3 = \{x \in \mathbb{L}^4 : \langle x, x\rangle = -1,\ x_0 > 0\}$$

is the *Minkowski model of (3-dimensional) hyperbolic space*; its tangent space at p is $T_p\mathbb{H}^3 = \{x \in \mathbb{L}^4 : \langle x, p\rangle = 0\}$.

Identifying $(x_0, x_1, x_2, x_3) \in \mathbb{L}^4$ with the Hermitian 2×2 matrix $\left(\begin{smallmatrix} x_0+x_3 & x_1+ix_2 \\ x_1-ix_2 & x_0-x_3 \end{smallmatrix}\right)$ realizes $\mathbb{L}^4$ as Herm(2), the space of Hermitian 2×2 matrices, with $\langle m, m\rangle = -\det(m)$; thus $\mathbb{H}^3 = \{m \in \mathrm{Herm}(2) : \det m = 1\}$. This matrix model is used throughout the construction of solutions.

For $p, u, v \in \mathbb{L}^4$ the *Lorentzian cross product* ("triple wedge product") is

$$p \wedge u \wedge v = -\det \begin{pmatrix} -i & j & k & l \\ p_0 & p_1 & p_2 & p_3 \\ u_0 & u_1 & u_2 & u_3 \\ v_0 & v_1 & v_2 & v_3 \end{pmatrix}, \tag{3}$$

(i, j, k, l) the standard basis, characterized by $\langle p \wedge u \wedge v, w\rangle = \det(p, u, v, w)$ for all w. Three identities proved directly from the determinant (identities (4)–(6) below; these are singled out by the author as small original computations used repeatedly in the proofs) are, for $\langle u, v\rangle = 0$, $w = p \wedge u \wedge v$, $\alpha = \langle u, u\rangle$, $\beta = \langle v, v\rangle$:

$$\langle w, w\rangle = -\alpha\beta\langle p, p\rangle + \alpha\langle p, v\rangle^2 + \beta\langle p, u\rangle^2, \tag{4}$$

$$\langle u \wedge v, u \wedge v\rangle = \langle u, u\rangle\langle v, v\rangle - \langle u, v\rangle^2 \qquad (u, v, w \in T_p\mathbb{H}^3), \tag{5}$$

$$w \wedge (u \wedge v) = u\langle w, v\rangle - v\langle w, u\rangle \qquad (u, v, w \in T_p\mathbb{H}^3). \tag{6}$$

The Lorentz metric restricted to each $T_p\mathbb{H}^3$ is positive definite, so $\mathbb{H}^3$ is Riemannian; by do Carmo [9] it is simply connected, complete, of constant sectional curvature -1. Note also: if $(\beta, v) \in \mathbb{H}^3 \times \mathbb{S}^3_1$ with $\langle \beta, v\rangle = 0$, then $n = \beta + v \in \mathbb{N}^3_+$ and $n_0 + n_3 \geq 0$.

2.2 Local theory of surfaces in $\mathbb{H}^3$

Let S be a connected, oriented 2-manifold (a *surface*) and $\psi : S \to \mathbb{H}^3$. A point z_0 is *regular* if $\mathrm{rank}(d\psi)_{z_0} = 2$; in a conformal parameter $z = s + it$ this is $\{\psi_s \wedge \psi_t\}_{z_0} \neq 0$. The *unit normal field* $\eta : S \to \mathbb{S}^3_1$ satisfies $\langle \psi, \eta\rangle = 0$, $\langle d\psi, \eta\rangle = 0$, and (matching the orientation) at regular points

$$\eta = \frac{\psi_s \wedge \psi_t}{\sqrt{\langle \psi_s \wedge \psi_t, \psi_s \wedge \psi_t\rangle}}. \tag{7}$$

At regular points there is an expansion

$$\eta_z = -\Delta\,\psi_z + \Upsilon\,\psi_{\bar z}, \qquad \eta_{\bar z} = \bar\Upsilon\,\psi_z - \Delta\,\psi_{\bar z}, \tag{8}$$

with Δ real-valued and Υ complex-valued; the mean curvature and Gauss–Kronecker curvature are $H = \Delta$ and $K = H^2 - |\Upsilon|^2$. Both are invariant under orientation-preserving reparametrization.

The *hyperbolic Gauss map* of ψ is $G = [\psi + \eta] : S \to \overline{\mathbb{C}}$,

$$G = [\psi + \eta] = \frac{N_1 - iN_2}{N_0 + N_3}, \qquad N = \psi + \eta, \tag{9}$$

i.e. the south-pole stereographic projection of the ideal boundary $\partial^2_\infty\mathbb{H}^3 \cong \overline{\mathbb{C}}$ onto the extended complex plane through the center of $\partial^2_\infty\mathbb{H}^3$ (with $G = \infty$ when $N_0 + N_3 = 0$). See Bryant [5] and Gálvez–Mira [2, 1] for details.

2.3 Riemann surfaces; meromorphic and anti-meromorphic maps

A *Riemann surface* is a 2-manifold with an atlas of holomorphically compatible charts. Every oriented Riemannian 2-manifold (X, σ) admits such an atlas in which σ is conformal (isothermal coordinates), so every oriented Riemannian surface may be regarded as a Riemann surface. The Riemann sphere $\partial^2_\infty \mathbb{H}^3 \cong \overline{\mathbb{C}}$ is an example.

A *meromorphic* function $f : S \to \overline{\mathbb{C}}$ is a holomorphic map in the manifold sense; equivalently, in an orientation-preserving conformal parameter z,

$$f_{\bar{z}} = 0. \tag{10}$$

Analogously f is *anti-meromorphic* if $f_z = 0$ in such parameters. Reversing the orientation of $\overline{\mathbb{C}}$ interchanges meromorphic and anti-meromorphic maps, so the two classes share properties such as uniqueness theorems and the identity theorem.

2.4 Conformality

A regular map $f : (M, \sigma_1) \to (N, \langle\cdot,\cdot\rangle)$ between Riemannian manifolds is *conformal* if σ_1 is a positive multiple of the pulled-back metric $\sigma_2(u, v) := \langle df(u), df(v)\rangle$. When $\dim M = 2$ and $z = s + it$ is σ_1-conformal, this is equivalent to

$$4\langle f_z, f_z\rangle = \langle f_s, f_s\rangle - \langle f_t, f_t\rangle - 2i\langle f_s, f_t\rangle = 0. \tag{11}$$

The notion extends to maps that are not everywhere regular and to targets with only a pseudo-metric, by requiring (11) in every conformal chart of M. A holomorphic map $f : D \subset \mathbb{C} \to \mathbb{C}$ is conformal at its regular points (angle- and orientation-preserving); conversely a conformal map $f : S \to \overline{\mathbb{C}}$ is holomorphic or anti-holomorphic, and vice versa.

2.5 Basic lemmas on BLW surfaces

Definition 1 (BLW surface)**.** A regular map $\psi : S \to \mathbb{H}^3$ is a **BLW surface** (Linear Weingarten surface of Bryant type) if there are real constants a, b such that

$$2a(H - 1) + bK_I = 2a(H - 1) + b(K - 1) = 0, \tag{12}$$

where H is the mean curvature, K the Gauss–Kronecker curvature, and $K_I = K - 1$ the (intrinsic) Gauss curvature of ψ.

The degenerate case $a + b = 0$ is treated in [6] and is excluded throughout ($a + b \neq 0$ is assumed from here on). If $b = 0$ then $a = \pm 1$ and (12) says $H \equiv 1$: these are the `CMC1` surfaces, for which the Björling problem was solved in [1]. If $a = 0$, (12) says $K \equiv 1$, i.e. the surface is *flat*, as studied in [3].

Lemma 2 ([2, Lemma 1])**.** *Let ψ be a BLW surface satisfying* (12)*. One may normalize $|a+b| = 1$, choosing the signs of a, b so that*

$$\sigma = aI + bII \tag{13}$$

is a positive-definite metric on S, where $I = \langle d\psi, d\psi\rangle$ and $II = \langle d\psi, -d\eta\rangle$ are the first and second fundamental forms.

Throughout, S is regarded as a Riemann surface for the conformal structure induced by σ, and it is always assumed $|a + b| = 1$ with signs chosen as in Lemma 2.

Remark 3 (assembled from [2, Thm. 1, its proof, and Remark 3]). Let $\psi : S \to \mathbb{H}^3$ be BLW as in (12), η its unit normal, z a σ-conformal parameter, and 2χ the conformal factor of $\sigma = 2\chi|dz|^2$. Then:

(1) $\psi+\eta$ is conformal for σ, and either $\psi(S)$ is part of a *horosphere*, or $\langle d(\psi+\eta), d(\psi+\eta)\rangle$ is a pseudo-metric of constant curvature $c=\frac{a}{a+b}$; in particular the hyperbolic Gauss map $G=[\psi+\eta]$ is conformal;

(2) $z_0\in S$ is a singular point of $\psi+\eta$ iff $H(z_0)=K(z_0)=1$; hence every such singular point is an *umbilic* point of ψ;

(3) the Hopf differential $Q_I=\langle\psi_z,\psi_z\rangle\,dz^2$ is holomorphic, so a BLW surface is either totally umbilic or has isolated umbilic points;

(4) at regular points of $\psi+\eta$,

$$\psi=\frac{2a+b}{2(a+b)}(\psi+\eta)+\frac{a+b}{\chi(2H-K-1)}(\psi+\eta)_{z\bar z}; \tag{14}$$

if in addition $b\neq 0$,

$$\frac{2H-K-1}{a+b}=\frac{1}{\chi}\langle(\psi+\eta)_z,(\psi+\eta)_{\bar z}\rangle, \tag{15}$$

and, with ϕ the conformal factor of $\langle d(\psi+\eta), d(\psi+\eta)\rangle$,

$$\psi=\frac{2a+b}{2(a+b)}(\psi+\eta)+\frac{2}{\phi}(\psi+\eta)_{z\bar z}. \tag{16}$$

By [1], formula (16) — the key device for *recovering* ψ from $\psi+\eta$ — remains valid even when $b=0$.

2.6 The Liouville equation

For a planar domain $U\subset\mathbb{C}$ and real constant c, the *Liouville equation* is

$$(\log\phi)_{z\bar z}=-\frac{c}{2}\phi. \tag{17}$$

If $ds^2=\phi|dz|^2$ is a Riemannian metric on U, then ϕ solves (17) iff ds^2 has constant curvature c. This link between surface theory and (17) is what drives the whole construction: solving the Björling problem for BLW surfaces reduces to solving a Cauchy problem for (17) (with $c=a/(a+b)$, the curvature of $\langle d(\psi+\eta),\cdot\rangle$ from Remark 3). This contrasts with the classical minimal-surface Björling problem in $\mathbb{R}^3$, which reduces merely to integrating functions with no real periods; here one must solve a genuine PDE.

The relevant Cauchy problem is

$$\begin{cases}(\log\phi)_{z\bar z}=-\dfrac{c}{2}\phi,\\ \phi(s,0)=a(s), \qquad \phi_z(s,0)=b(s),\end{cases} \tag{18}$$

with $a(s)$ real-analytic and non-negative and $b(s)$ complex-analytic with $2\Re\{b(s)\}=a'(s)$. Write $\{f,z\}=\big(f''/f'\big)'-\frac{1}{2}\big(f''/f'\big)^2$ for the Schwarzian derivative.

Theorem 4 ([1, Thm. 1], case $c\neq 0$)**.** *Let $c\neq 0$ and let $T(s):I\to\mathbb{C}$ solve $\{T,s\}=\Upsilon(s)$, where*

$$2\Upsilon(s)=2\Big(\frac{b(s)}{a(s)}\Big)'-\Big(\frac{b(s)}{a(s)}\Big)^2+c\,a(s).$$

Set

$$\overline{R}(s)=\frac{1}{c}\,\frac{T''(s)-\{b(s)/a(s)\}\,T'(s)}{2T'(s)^2-T(s)T''(s)+\{b(s)/a(s)\}\,T(s)T'(s)}.$$

Let $T(z)$, $R(z)$ be the meromorphic extensions of $T(s)$, $\overline{R}(s)$ on an open $D \subset \mathbb{C}$ containing I. Then

$$\phi(s,t) = \frac{4\,T_z\,\overline{R}_{\bar{z}}}{(1+cT\overline{R})^2}$$

is the unique solution of (18).

For the degenerate case $c = 0$ the Cauchy data is instead $\phi(s,0) = a(s)$, $\phi_t(s,0) = d(s)$ (real-analytic $a > 0$, d):

Theorem 5 ([1, Thm. 6], case $c = 0$)**.** *Fix $s_0 \in I$, put $\theta(s) = -\frac{1}{2}\int_{s_0}^{s} d(r)/a(r)\,dr$, and let $\sqrt{a}(z)$, $\theta(z)$ be the holomorphic extensions of $\sqrt{a(s)}$, $\theta(s)$. Then $\phi(s,t) = |f(z)|^2$, $z = s + it$, with*

$$f(z) = \sqrt{a}(z)\,e^{i\theta(z)}.$$

There is also a purely *geometric* construction of the solution to the real form of (18),

$$\Delta(\ln\phi) = -2c\phi, \qquad \phi(s,0) = a(s),\ \phi_t(s,0) = d(s). \tag{19}$$

Let $\mathfrak{Q}(c)$ be the constant-curvature-c Riemann surface $\mathfrak{Q}(0) = \mathbb{R}^2$, $\mathfrak{Q}(c) = \mathbb{S}^2(c)$ ($c > 0$) or $\mathbb{H}^2(c)$ ($c < 0$), and $\pi(x_0, x_1, x_2) = \frac{x_1 + ix_2}{1 - cx_0}$ its stereographic projection to $\overline{\mathbb{C}}$.

Theorem 6 ([1, Thm. 3])**.** *The unique solution of* (19) *is*

$$\phi(s,t) = \frac{4|g'(z)|^2}{(1+c|g(z)|^2)^2}, \qquad z = s + it,$$

where $g(z)$ is the meromorphic extension of $g(s) = \pi(\alpha(s))$, and $\alpha(s)$ is the unique arclength-parametrized curve in $\mathfrak{Q}(c)$ with geodesic curvature $\kappa(s) = -d(s)/(2a(s)^{3/2})$ and speed function $u(s) = \int^s \sqrt{a}(r)\,dr$.

Finally, the *modified Liouville equation* used directly in the Björling construction is

$$4(\ln\rho)_{z\bar{z}} = -c\,\rho^2\,|f(z)|^2, \tag{20}$$

f meromorphic: ρ solves (20) iff $\phi = \rho^2|f|^2$ solves (17). For its Cauchy problem $\rho(s,0) = v(s)$, $\rho_z(s,0) = w(s)$ (f with no real pole):

Corollary 7. *The solution of the Cauchy problem for* (20) *is $\rho = \sqrt{\phi}/|f|$, with ϕ the solution of* (18) *(via Theorem 4 or 5) with*

$$a(s) = v(s)^2|f(s)|^2, \qquad b(s) = 2v(s)w(s)|f(s)|^2 + v(s)^2 f'(s)\overline{f(s)}.$$

3 The Björling problem for BLW surfaces

Definition 8. A curve $\beta : I \to S$ on a Riemann surface S is *analytic at $s_0 \in I$* if, in a conformal chart (U, z) around $\beta(s_0)$, the curve $z \circ \beta$ is real-analytic at s_0; β is *analytic* if this holds at every $s \in I$. Regularity of β is defined analogously.

3.1 Statement of the problem

Problem 9 (Björling problem for BLW surfaces)**.** Let $a, b \in \mathbb{R}$ with $|a + b| = 1$, let $\beta : I \to \mathbb{H}^3$ be a regular analytic curve and $V : I \to \mathbb{S}^3_1$ an analytic vector field along β such that, for all $s \in I$,

$$\langle \beta(s), V(s)\rangle = \langle \beta'(s), V(s)\rangle = 0, \tag{21}$$

$$\langle \beta'(s), a\beta'(s) - bV'(s)\rangle > 0. \tag{22}$$

Find a BLW surface $\psi : S \to \mathbb{H}^3$ satisfying (12) such that there is a regular analytic curve $\Gamma \subset S$ with $\psi(\Gamma) = \beta$ and with unit normal field of ψ along β equal to V. Such a ψ is a *solution of the Björling problem* with *initial data* (β, V).

Condition (22) is exactly the requirement that the induced conformal factor χ of $\sigma = aI + bII$ be positive along β (see (32) below), i.e. that σ have a chance of being positive-definite near β.

3.2 Main theorem

Theorem 10 (Existence and uniqueness)**.** *Let (β, V) be initial data as in Problem 9. Suppose $v = \beta + V$ satisfies $v_0(s) + v_3(s) \neq 0$ for all $s \in I$, and that the map*

$$G = \frac{v_1 - iv_2}{v_0 + v_3} : I \to \mathbb{C} \tag{23}$$

has nowhere-vanishing derivative on I. Then the Björling problem with initial data (β, V) has a unique *solution $\psi : D \subset \mathbb{C} \to \mathbb{H}^3$, constructed near β as follows.*

Case $a + b = 1$**.** *Let $\beta(z)$, $V(z)$ be the holomorphic extensions of $\beta(s)$, $V(s)$; set $v(z) = \beta(z) + V(z)$ and*

$$G(z) = \frac{v_1(z) - iv_2(z)}{v_0(z) + v_3(z)}. \tag{24}$$

Let $\rho : D \to \mathbb{C} \setminus \{0\}$ solve the Cauchy problem

$$\begin{cases} 4(\log\rho)_{z\bar{z}} = -\dfrac{a}{a+b}\,\rho^2|G_z|^2, \\ \rho(s,0) = v_0(s) + v_3(s), \\ \rho_z(s,0) = \dfrac{1}{2}\Big\{v_0' + v_3' + i\big[(V \wedge v')_0 + (V \wedge v')_3\big]\Big\}; \end{cases} \tag{25}$$

then $\psi = F\Omega F^* : D \to \mathbb{H}^3$, where

$$F = \begin{pmatrix} 1 & 0 \\ G & 1 \end{pmatrix}, \qquad \Omega = \begin{pmatrix} \dfrac{2a+b}{2(a+b)}\rho + \dfrac{2\rho_{z\bar{z}}}{\rho^2|G_z|^2} & \dfrac{2\rho_z}{\rho^2 G_z} \\ \dfrac{2\rho_{\bar{z}}}{\rho^2\overline{G_z}} & \dfrac{2}{\rho} \end{pmatrix}. \tag{26}$$

Case $a + b = -1$**.** *Let $\beta(z)$, $V(z)$ be the* anti-*holomorphic extensions of $\beta(s)$, $V(s)$; set $v(z) = \beta(z) + V(z)$ and $G(z)$ as in (24). Let $\rho : D \to \mathbb{C} \setminus \{0\}$ solve*

$$\begin{cases} 4(\log\rho)_{z\bar{z}} = -\dfrac{a}{a+b}\,\rho^2|\overline{G_z}|^2, \\ \rho(s,0) = v_0(s) + v_3(s), \\ \rho_z(s,0) = \dfrac{1}{2}\Big\{v_0' + v_3' - i\big[(V \wedge v')_0 + (V \wedge v')_3\big]\Big\}; \end{cases} \tag{27}$$

then $\psi = F\Omega F^* : D \to \mathbb{H}^3$ with F as before and

$$\Omega = \begin{pmatrix} \dfrac{2a+b}{2(a+b)}\rho + \dfrac{2\rho_{z\bar{z}}}{\rho^2|\overline{G_z}|^2} & \dfrac{2\rho_{\bar{z}}}{\rho^2 G_{\bar{z}}} \\ \dfrac{2\rho_z}{\rho^2\overline{G_z}} & \dfrac{2}{\rho} \end{pmatrix}. \tag{28}$$

In either case, G is the hyperbolic Gauss map of ψ, and ρ is constructed via Section 2.6.

Two remarks on the hypotheses (elaborated in Section 3.5): the non-vanishing of $v_0 + v_3$ can be removed (composing G with a Möbius transformation), while the non-vanishing of G' genuinely limits singularities and cannot be dropped for free (Section 4, Problems 4., 6.).

The theorem *contains* the Björling theorems for `CMC1` surfaces [1] ($b = 0$) and regular flat surfaces [3] ($a = 0$) as special cases, and gives a *Björling representation* of BLW surfaces that bypasses the conformal representation of [2].

3.3 Meromorphicity of the hyperbolic Gauss map

By Bryant [5], the hyperbolic Gauss map of a regular surface $\psi : S \to \mathbb{H}^3$ is meromorphic if and only if ψ is `CMC1`. The following theorem partially generalizes this fact and is the key technical tool used to justify meromorphic vs. anti-meromorphic extension of G in the two cases of Theorem 10.

Theorem 11. *Let $\psi : S \to \mathbb{H}^3$ be a BLW surface satisfying (12), with $\psi(S)$ not (part of) a horosphere. Then the hyperbolic Gauss map G of ψ is meromorphic if $a + b > 0$, and anti-meromorphic if $a + b < 0$.*

For `CMC1` surfaces $a+b = a = \pm 1$: Bryant's theorem is the special case $a = 1$ (meromorphic); the sign convention in the BLW normalization absorbs the case $a = -1$ as the anti-meromorphic statement, consistent with the classical result up to orientation.

3.4 Sketch of proof

The proof of Theorem 10 is organized into a uniqueness part (Section 3.4.1) and an existence part (Section 3.4.2) for the case $a + b = 1$; the case $a + b = -1$ (Section 3.4.3) is handled by an entirely analogous argument. Section 3.5 proves Theorem 11 and discusses the hypotheses on the initial data.

3.4.1 Uniqueness ($a + b = 1$)

Suppose $\psi : S \to \mathbb{H}^3$ solves Problem 9. Equip S with the Riemann surface structure induced by $\sigma = aI + bII$ (Lemma 2). The strategy is to show that ψ is *locally determined* by the initial data (β, V) via an explicit formula, and then invoke analytic continuation to get global uniqueness.

Parametrization. Choose a conformal chart (U, z) meeting Γ in a segment J'; write $\gamma = z \circ \Gamma$. Since γ is analytic and regular at a fixed $s_0 \in J'$, the inverse function theorem gives a biholomorphism $\gamma : D \to B$ onto a neighborhood in $z(U)$, and the surface is represented locally as $\psi(w) = \chi(\gamma(w))$ with $\psi(s, 0) = \beta(s)$, $\eta(s, 0) = V(s)$ on $J = (s_0 - \delta, s_0 + \delta)$.

Reduction to the modified Liouville equation. Writing $v = \beta + V : I \to \mathbb{N}^3_+$, the hyperbolic Gauss map restricted to β is $G(s) = (v_1 - iv_2)/(v_0 + v_3)$; since G is meromorphic (justified in Section 3.5) it extends uniquely to $G(z)$ on D. Writing

$$\psi + \eta = \rho \begin{pmatrix} 1 & \overline{G} \\ G & G\overline{G} \end{pmatrix}, \qquad \rho = N_0 + N_3,\ N = \psi + \eta, \tag{29}$$

the fact that $\tau = \langle d(\psi + \eta), d(\psi + \eta) \rangle$ has constant curvature $c = a/(a + b)$ (Remark 3) together with (29) gives $\langle (\psi + \eta)_z, (\psi + \eta)_{\bar{z}} \rangle = \frac{1}{2}\rho^2 |G_z|^2$, so the conformal factor $\phi = \rho^2 |G_z|^2$ of τ satisfies the Liouville equation (17); since G_z is meromorphic, ρ itself satisfies the *modified* Liouville equation (20) with $c = a/(a + b)$.

Initial conditions. Directly, $\rho(s, 0) = v_0(s) + v_3(s)$. To find $\rho_z(s, 0)$, expand $\eta_z, \eta_{\bar{z}}$ via (8) to get $(\psi + \eta)_z \wedge (\psi + \eta)_{\bar{z}} = -(2H - K - 1)\psi_z \wedge \psi_{\bar{z}}$, and combine with (15) (valid, by Bryant/[1], even when $b = 0$) to get $\operatorname{sgn}(2H - K - 1) = \operatorname{sgn}(a + b)$ at regular points, hence

$$(\psi + \eta)_t = -\operatorname{sgn}(a + b)\, \eta \wedge (\psi + \eta)_s.$$

This yields $(\psi + \eta)_z(s, 0) = \frac{1}{2}\{v' + i \operatorname{sgn}(a + b)\, V \wedge v'\}$, i.e.

$$\rho_z(s, 0) = \frac{1}{2}\Big\{ v'_0 + v'_3 + i \operatorname{sgn}(a + b) \big[(V \wedge v')_0 + (V \wedge v')_3 \big] \Big\}, \tag{30}$$

which together with the previous two facts is exactly the Cauchy problem (25) (for $a + b = 1$; sign flips to (27) when $a + b = -1$). Thus ρ is uniquely determined by (β, V).

Recovering ψ. With $F = \left(\begin{smallmatrix} 1 & 0 \\ G & 1 \end{smallmatrix}\right)$ one has $\psi + \eta = F\left(\begin{smallmatrix} \rho & 0 \\ 0 & 0 \end{smallmatrix}\right)F^*$, and differentiating using $F_z = F\left(\begin{smallmatrix} 0 & 0 \\ G_z & 0 \end{smallmatrix}\right)$ gives explicit expressions for $(\psi+\eta)_{\bar{z}}$ and $(\psi+\eta)_{z\bar{z}}$ in terms of F and ρ. Substituting into the recovery formula (16) from Remark 3 gives, at regular points of $\psi + \eta$,

$$\psi = F\Omega F^*$$

with Ω exactly (26), and $\det\Omega = 1$ because ρ solves (25). This is the desired closed form, completing the uniqueness argument (via analytic continuation of BLW surfaces).

3.4.2 Existence ($a + b = 1$)

Given data (β, V) satisfying the hypotheses of Theorem 10 with $a+b=1$: define $G(s)$ by (23), extend it meromorphically to $G(z)$ by (24) on a domain D containing I, and let ρ solve (25). On $U = D \setminus P$ (removing the closed set P of zeros of ρ and zeros/poles of G_z — a set that does not meet I), define $\psi = F\Omega F^*$ and $N = F\left(\begin{smallmatrix} \rho & 0 \\ 0 & 0 \end{smallmatrix}\right)F^*$ with F, Ω as in (26). The proof that ψ is a solution proceeds in five steps.

(i) $\sigma = aI + bII$ **is positive-definite.** Writing $\Omega = \left(\begin{smallmatrix} A & B \\ \overline{B} & C \end{smallmatrix}\right)$ and $U := AG_z + \overline{B}_z = \frac{b}{2(a+b)}\rho G_z$ (using (25)), direct computation with $F_z = F\left(\begin{smallmatrix} 0 & 0 \\ G_z & 0 \end{smallmatrix}\right)$ shows the dz^2- and $d\bar{z}^2$-coefficients of σ vanish and the remaining ($|dz|^2$) coefficient is

$$2\chi = (a+b)\big\{|B_z|^2 - |U|^2\big\}. \tag{31}$$

Along β this equals

$$2\chi(s,0) = 2\langle \beta'(s), a\beta'(s) - bV'(s)\rangle > 0 \tag{32}$$

by hypothesis (22), so σ is positive-definite (hence a genuine conformal metric) near β.

(ii) ψ **is regular.** From $\psi_z = F\left(\begin{smallmatrix} * & B_z \\ U & 0 \end{smallmatrix}\right)F^*$ one computes $\langle \psi_s \wedge \psi_t, \psi_s \wedge \psi_t\rangle = |B_z|^2 - |U|^2$, which is exactly the (positive, by step (i)) coefficient of σ; hence $\psi_s \wedge \psi_t \neq 0$ near β.

(iii) $\eta := N - \psi$ **is the unit normal.** Using $\det\Omega = 1$ and $\rho C = 2$ one checks $\langle \eta, \eta\rangle = 1$ and $\langle \psi, \eta\rangle = 0$ by direct matrix-determinant identities; finally $\langle \psi_z \wedge \psi_{\bar{z}}, \eta\rangle = \frac{i}{2}\{|B_z|^2 - |U|^2\}$ (computed via the symbolic routines of Appendix B), and since $a + b = 1$, (31) forces $\langle \psi_s \wedge \psi_t, \eta\rangle > 0$, i.e. η has the correct orientation.

(iv) ψ **is BLW.** Substituting the computed $d\eta$ into (8) gives a 2×2 linear system for H, Υ with nonzero determinant $D = |U|^2 - |B_z|^2$, whose solution is

$$H = 1 - \frac{b}{2(a+b)}\frac{\rho^2|G_z|^2}{D}, \qquad K = H^2 - |\Upsilon|^2 = 1 + \frac{a}{a+b}\frac{\rho^2|G_z|^2}{D},$$

from which $2a(H-1) + b(K-1) = 0$ holds identically.

(v) ψ **realizes the data** (β, V)**.** An identity relating the initial curve data,

$$(v_0+v_3)^2 G'\big\{\beta_1 + i\beta_2 - \overline{G}(\beta_0+\beta_3)\big\} = v_0' + v_3' + i\big[(V\wedge v')_0 + (V\wedge v')_3\big], \tag{33}$$

combined with the third line of (25), gives $B(s,0) = \beta_1 + i\beta_2 - \overline{G}(\beta_0+\beta_3)$, and then ($\det\Omega = 1$, $\rho C = 2$) $A(s,0) = \beta_0 + \beta_3$. Writing out $\psi = F\Omega F^*$ entrywise along β and using $\langle \beta, \beta\rangle = -1$ (so $\beta_0 + \beta_3 \neq 0$) pins down the remaining matrix entry, giving $\psi(s,0) = \beta(s)$ for all $s \in I$; since $(\psi+\eta)(s,0) = v(s) = \beta(s) + V(s)$ by construction, this also gives $\eta(s,0) = V(s)$.

Together, (i)–(v) show ψ solves the Björling problem with data (β, V), completing the existence proof for $a + b = 1$.

3.4.3 The case $a + b = -1$

The argument mirrors Sections 3.4.1–3.4.2 verbatim, with G now extended *anti*-meromorphically (justified by Theorem 11 for $a + b < 0$). The representation (29) for $\psi + \eta$ is unchanged, but now $\langle (\psi+\eta)_z, (\psi+\eta)_{\bar{z}} \rangle = \frac{1}{2}\rho^2 |\overline{G_z}|^2$, giving the modified Liouville equation with $\overline{G_z}$ in place of G_z, and Cauchy data (27) (a sign flip in the imaginary part relative to (25), coming from $\operatorname{sgn}(a+b) = -1$ in (30)). The recovery formula gives $\psi = F\Omega F^*$ with Ω as in (28). All five existence steps go through with the roles of U, B_z swapped appropriately, giving

$$H = 1 - \frac{b}{2(a+b)} \frac{\rho^2 |\overline{G_z}|^2}{D}, \qquad K = 1 + \frac{a}{a+b} \frac{\rho^2 |\overline{G_z}|^2}{D}, \qquad D = |U|^2 - |\overline{B_z}|^2 \neq 0,$$

and $\psi(s, 0) = \beta(s)$ follows exactly as before.

3.5 Further remarks

3.5.1 On the hypotheses on the initial data

Writing $Q = a\psi - b\eta$ and using $\sigma = \langle d\psi, dQ \rangle = 2\chi |dz|^2$ in a σ-conformal chart, one gets $2\chi(s, 0) = \langle \beta'(s), a\beta'(s) - bV'(s) \rangle > 0$: this is exactly why condition (22) is imposed in the statement of the problem — it is a *necessary* condition for any solution to exist, not an artificial restriction.

The hypothesis $v_0(s) + v_3(s) \neq 0$ (needed for (24) to make sense) can be dropped following [1]: composing G with a suitable Möbius transformation of $\overline{\mathbb{C}}$ reduces to this case. Concretely, since $v(s) = \beta(s) + V(s) \neq 0$, whenever $v_0 + v_3 = 0$ one has $v_0 - v_3 \neq 0$, and one instead writes $N = \psi + \eta = \rho_* \left(\begin{smallmatrix} H\overline{H} & \overline{H} \\ H & 1 \end{smallmatrix} \right)$, $\rho_* = N_0 - N_3$, $H = (N_1 - iN_2)/(N_0 - N_3)$, with the extra requirement $H'(s_0) \neq 0$ at the exceptional point; the same computations then give the same conclusion of Theorem 10 by locality of the problem.

The nowhere-vanishing of G' is more essential. If $G' \equiv 0$ identically, Remark 3 forces any solution to be (part of) a horosphere. If $G' \not\equiv 0$ but has isolated zeros (forced to be isolated by analyticity), the construction of Theorem 10 can develop singularities; analyzing these singularities is left as an open problem (Section 4, Problems 4., 6.).

3.5.2 Why G is meromorphic: proof of Theorem 11

By Remark 3(1), G is always conformal, hence meromorphic or anti-meromorphic; the content of Theorem 11 is pinning down *which one*, by the sign of $a + b$. The uniqueness argument of Section 3.4.1 tacitly assumes G meromorphic for $a + b = 1$; this is justified by contradiction: assume ψ is BLW, $a + b = 1$, σ positive-definite, $\psi(S)$ not a horosphere (so the umbilic points — equivalently the zeros of $G_{\bar{z}}$ — are isolated), and suppose *towards contradiction* that G is anti-meromorphic. Fix a conformal chart (U, z) (preserving orientation) and write $\psi + \eta$ as in (29) with ρ nonvanishing on U (achievable after composing G with a suitable Möbius map). The computation of Section 3.4.3 (the $a + b = -1$ formulas, which apply here because they only used anti-meromorphicity of G, not the sign of $a + b$) gives $\psi = F\Omega F^*$ with Ω as in (28), hence (31) for $a + b = 1$ forces $|\overline{B_z}|^2 - |U|^2 > 0$; but then the analogue of the identity used in step (iii) of Section 3.4.2 gives $\langle \psi_s \wedge \psi_t, \eta \rangle < 0$ — contradicting that η and $\psi_s \wedge \psi_t$ point in the same direction (by construction of the unit normal, Section 2.2). Hence G must be meromorphic when $a + b = 1$; the case $a + b = -1$ (anti-meromorphic) is symmetric.

4 Conclusions and open problems

The thesis poses and solves the Björling problem for BLW surfaces. Theorem 10 establishes existence and uniqueness of the solution; its representation formula does not use the conformal representation of [2], and is thus a genuine *Björling representation* for BLW surfaces. It

specializes to the known results for `CMC1` surfaces [1] and regular flat surfaces [3]. Theorem 11 determines the meromorphicity type of the hyperbolic Gauss map of a BLW surface from the sign of $a + b$. Along the way, the author establishes the Lorentzian cross-product identities (4)–(6), which are used repeatedly and are noted as small original computational results.

For lack of time, the thesis does not address one of the important consequences of solving the Björling problem: the *symmetry principle* for BLW surfaces. This should follow by an argument entirely parallel to [1] once an appropriate notion of symmetry (for BLW surfaces reflected across geodesic planes orthogonal to them) is set up.

Open problems

The following related problems (some already studied for `CMC1` surfaces in [1], as noted) are proposed:

1. Study BLW surfaces with the topology of a cylinder.

2. Study complete BLW surfaces with special finite *dual* total curvature.

3. Find an explicit closed formula for the solution when the initial curve β is planar.

4. Solve the Björling problem when G in (23) is a *constant* map; as noted in Section 3.5, any solution in this case must be (part of) a horosphere.

5. Find a geometric solution of the Cauchy problem for the Liouville equation (19) (with $c = 1$) directly, in the spirit of [1]'s geometric approach via the secondary Gauss map (related to the rigidity property) of `CMC1` surfaces, recovering the solution of (19) from that map and the initial data.

6. Solve the Björling problem allowing singularities — e.g. where G' or $\langle \beta', a\beta' - bV' \rangle$ vanish at isolated points of I — extending the notion of BLW surface to points where (12) only holds at regular points. This parallels [3]'s treatment of flat fronts (which allow singularities, unlike flat surfaces) and its subsequent classification of flat fronts with embedded isolated singularities.

7. Solve the Björling problem when the initial data satisfies $\langle \beta', a\beta' - bV' \rangle \equiv 0$ identically (note: by analyticity of β, V, the zero set of this quantity is either isolated or all of I). By Corollary 10 of [3] (case $a = 0$), given β and a suitably chosen V, there is a unique flat front Σ for which β consists exactly of the singular points of Σ.

8. Solve the Björling problem for `ELW` (Elliptic Linear Weingarten) surfaces [4] — surfaces in $\mathbb{R}^3$ satisfying an analogous relation between H and K. This case may in fact be *harder* than the BLW case: for `ELW` surfaces only a *harmonic*, not conformal, representation is available [4], which is the chief obstacle.

9. By [5, 1], the hyperbolic Gauss map of a regular surface $\psi : S \to \mathbb{H}^3$ is meromorphic iff ψ is `CMC1`; Theorem 11 partially generalizes this. The converse direction is open: if $\sigma = aI + bII$ is positive-definite and G is meromorphic (or anti-meromorphic), must ψ be BLW?

Finally, there is also a Björling problem for *maximal* surfaces in $\mathbb{L}^n$; preliminary computations (Assoc. Prof. Đoàn Thế Hiếu, based on [12, 11]) suggest this can be solved by integrating functions with no real periods.

A Previously solved Björling-type problems

The following table (reproduced from the thesis's Appendix A) lists Björling-type problems solved prior to this thesis. In every case the solution is unique and yields a symmetry principle for the surface class considered.

Surface class	*Reference*	*Author(s)*	*Year*
minimal, $\mathbb{R}^3$	[10]	E. G. Björling, H. A. Schwarz	1844, 1890
minimal, $\mathbb{R}^n$	[11]	L. J. Alías, P. Mira	2002
maximal, $\mathbb{L}^3$	[12]	L. J. Alías et al.	2003
minimal Möbius strip, $\mathbb{R}^n$	[13]	P. Mira	2005
flat front, $\mathbb{H}^3$	[3]	J. A. Gálvez, P. Mira	2005
flat surface, $\mathbb{H}^3$	[3]	J. A. Gálvez, P. Mira	2005
`CMC1`, $\mathbb{H}^3$	[1]	J. A. Gálvez, P. Mira	2005
improper affine sphere, $\mathbb{R}^3$	[14]	J. A. Aledo et al.	(preprint)
BLW, $\mathbb{H}^3$	*this thesis*	Huỳnh Kỳ Anh	2006

B Note on the appendix on symbolic computation

The thesis includes an appendix with Maple routines used to verify, by symbolic (Hermitian 2×2 matrix) computation, identities such as $\langle \psi_z \wedge \psi_{\bar{z}}, \eta \rangle = \frac{i}{2}\{|B_z|^2 - |U|^2\}$ (used in step (iii) of Section 3.4.2) for the case $a + b = 1$, and the analogous identity with $U, \overline{B_z}$ swapped for $a+b = -1$ — that are otherwise long, purely mechanical determinant expansions. These routines are a computational aid, not a separate mathematical contribution, and are not reproduced here.

References

[1] J. A. Gálvez, A. Martínez, P. Mira, *The Björling problem for constant mean curvature one surfaces in the hyperbolic space*, related works of Gálvez–Mira on `CMC1` surfaces.

[2] J. A. Gálvez, A. Martínez, P. Mira, *On the conformal (harmonic) representation of linear Weingarten surfaces of Bryant type in* $\mathbb{H}^3$ (referenced as a preprint at the time of writing).

[3] J. A. Gálvez, P. Mira, *Embedded isolated singularities of flat surfaces in hyperbolic space* and the Björling problem for flat fronts/flat surfaces, 2005.

[4] J. A. Gálvez, A. Martínez, P. Mira, on Elliptic Linear Weingarten (`ELW`) surfaces in $\mathbb{R}^3$.

[5] R. Bryant, *Surfaces of mean curvature one in hyperbolic space*, Astérisque **154–155** (1987).

[6] Reference on the degenerate BLW case $a + b = 0$ (cited via [2]; not available to the author of the thesis).

[7] Local theory of surfaces reference (Fujimori–Kobayashi–Rossman or similar).

[8] S. Fujimori, concise treatment of the basic equations of surface theory.

[9] M. P. do Carmo, treatment of the hyperbolic space model with detailed proofs.

[10] K. Fung, treatment of minimal surfaces in $\mathbb{R}^3$ as isotropic curves in $\mathbb{C}^3$, with a fast solution of the classical Björling problem.

[11] L. J. Alías, P. Mira, Björling problem for minimal surfaces in $\mathbb{R}^n$, 2002.

[12] L. J. Alías et al., Björling problem for maximal surfaces in $\mathbb{L}^3$, 2003.

[13] P. Mira, Björling problem for minimal Möbius strips in $\mathbb{R}^n$, 2005.

[14] J. A. Aledo et al., Björling problem for improper affine spheres in $\mathbb{R}^3$ (preprint).

[15] Reference(s) on the Liouville equation and its Cauchy problem.

[16] N. Hitchin, lecture notes on Riemann surfaces.

[17] Wikipedia, *Riemann surface*.

[18] Wikipedia, *Analytic continuation*.

[19] L. Ahlfors, *Complex Analysis*.